# Worksheet 2.3 — Composite and Inverse Functions

Mathematics: Analysis & Approaches SL (IB) · Unit 2: Functions

### The Big Ideas

Composite functions chain two functions; inverse functions undo a function completely.

- $(f \circ g)(x) = f(g(x))$ — evaluate the inside function first.
- To find $f^{-1}$: swap $x$ and $y$ in $y = f(x)$, then solve for $y$.
- $f$ and $f^{-1}$ are reflections of each other in $y = x$; $f$ needs a restricted domain to have an inverse if it fails the horizontal line test.

## Worked Examples

### Example 1 — Evaluate a composite

Given $f(x) = 3x - 2$, $g(x) = x^2 + 1$, find $f(g(-1))$.

***Solution.*** $g(-1) = 1 + 1 = 2$. $f(2) = 3(2) - 2$.

**Conclusion:** $f(g(-1)) = 4$.

### Example 2 — Find a composite expression

Given $f(x) = 3x - 2$, $g(x) = x^2 + 1$, find $(f \circ g)(x)$.

***Solution.*** $f(g(x)) = 3(x^2 + 1) - 2 = 3x^2 + 3 - 2$.

**Conclusion:** $(f \circ g)(x) = 3x^2 + 1$.

### Example 3 — Find an inverse

Find the inverse of $f(x) = 4x + 7$.

***Solution.*** $y = 4x + 7$; swap: $x = 4y + 7 \Rightarrow y = \dfrac{x - 7}{4}$.

**Conclusion:** $f^{-1}(x) = \dfrac{x - 7}{4}$.

### Example 4 — Inverse of a restricted quadratic

Find the inverse of $f(x) = x^2 - 3$, restricted to $x \geq 0$.

***Solution.*** $y = x^2 - 3$; swap: $x = y^2 - 3 \Rightarrow y^2 = x + 3 \Rightarrow y = \sqrt{x + 3}$ (positive root, matching the restriction).

**Conclusion:** $f^{-1}(x) = \sqrt{x + 3}$.

### Example 5 — Composite requiring domain awareness

Given $f(x) = \sqrt{x}$, $g(x) = x - 5$, find the domain of $(f \circ g)(x)$.

***Solution.*** $(f \circ g)(x) = \sqrt{x - 5}$, which requires $x - 5 \geq 0$.

**Conclusion:** domain is $x \geq 5$.

### Example 6 — Finding a function given its composite

Given $g(x) = 2x + 1$ and $(f \circ g)(x) = 4x^2 + 4x + 3$, find $f(x)$.

***Solution.*** Since $(f \circ g)(x) = f(2x+1)$, let $u = 2x+1 \Rightarrow x = \dfrac{u-1}{2}$. Substitute into $4x^2+4x+3$: after simplifying, this equals $u^2 + 2$.

**Conclusion:** $f(x) = x^2 + 2$. (Check: $f(2x+1) = (2x+1)^2 + 2 = 4x^2 + 4x + 1 + 2 = 4x^2 + 4x + 3$ )

### Example 7 — Inverse of a rational function

Find the inverse of $f(x) = \dfrac{2x+1}{x-3}$.

***Solution.*** $y = \dfrac{2x+1}{x-3}$; swap: $x = \dfrac{2y+1}{y-3} \Rightarrow x(y-3) = 2y+1 \Rightarrow xy - 3x = 2y + 1$.

$xy - 2y = 3x + 1 \Rightarrow y(x-2) = 3x+1 \Rightarrow y = \dfrac{3x+1}{x-2}$.

**Conclusion:** $f^{-1}(x) = \dfrac{3x+1}{x-2}$.

### Example 8 — Composing a function with its own inverse

Given $f(x) = \dfrac{x-4}{2}$, verify that $f(f^{-1}(10)) = 10$ without finding $f^{-1}$'s formula.

***Solution.*** By definition, applying $f$ after $f^{-1}$ always returns the original input.

**Conclusion:** $f(f^{-1}(10)) = 10$ directly, no formula needed.

### Example 9 — Domain restriction needed for an inverse to exist

Explain why $f(x) = (x-2)^2 + 1$ needs a restricted domain to have an inverse that is a function, and state a valid restriction.

***Solution.*** Without restriction, $f$ fails the horizontal line test (e.g. $f(1) = f(3) = 2$).

**Conclusion:** restricting to $x \geq 2$ (or $x \leq 2$) makes $f$ one-to-one, so its inverse is then a function.

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

Given $f(x) = x + 6$, $g(x) = 2x$, find $g(f(2))$.

### Question 2

Given $f(x) = x + 6$, $g(x) = 2x$, find $(g \circ f)(x)$.

### Question 3

Find the inverse of $f(x) = 5x - 3$.

### Question 4

Find the inverse of $f(x) = \dfrac{x + 4}{3}$.

### Question 5

Find the inverse of $f(x) = x^2 + 2$, restricted to $x \leq 0$.

## Question 6

Verify that $f(x) = 2x + 6$ and $g(x) = \dfrac{x - 6}{2}$ are inverses by computing both compositions.

## Question 7

Given $f(x) = x - 1$, $g(x) = x^2$, find $(g \circ f)(3)$ and $(f \circ g)(3)$, and note they differ.

## Question 8

Given $f(x) = \sqrt{x}$, $g(x) = x + 8$, find the domain of $(f \circ g)(x)$.

## Question 9

Find the inverse of $f(x) = \dfrac{3x - 2}{x + 1}$.

## Question 10

Explain why $f(x) = x^2 - 6x + 5$ needs a restricted domain to have a function inverse, and state a valid restriction.

## Question 11 — Challenge

Given $f(x) = 2x - 5$, find $f^{-1}(f(8))$ without computing $f^{-1}$'s formula, and explain why your shortcut works.

## Question 12 — Challenge

Given $g(x) = 3x - 2$ and $(f \circ g)(x) = 9x^2 - 12x + 7$, find $f(x)$.

## Question 13 — Challenge

Given $f(x) = \dfrac{x}{x-1}$, find $f^{-1}(x)$ and verify $f^{-1}$ has the same rule as $f$ (i.e. $f$ is its own inverse).

# Question Recap & Answer Key

## Question Recap

1. Given $f(x) = x + 6$, $g(x) = 2x$, find $g(f(2))$.
2. Given $f(x) = x + 6$, $g(x) = 2x$, find $(g \circ f)(x)$.
3. Find the inverse of $f(x) = 5x - 3$.
4. Find the inverse of $f(x) = \dfrac{x+4}{3}$.
5. Find the inverse of $f(x) = x^2 + 2$, restricted to $x \leq 0$.
6. Verify that $f(x) = 2x + 6$ and $g(x) = \dfrac{x-6}{2}$ are inverses by computing both compositions.
7. Given $f(x) = x - 1$, $g(x) = x^2$, find $(g \circ f)(3)$ and $(f \circ g)(3)$, and note they differ.
8. Given $f(x) = \sqrt{x}$, $g(x) = x + 8$, find the domain of $(f \circ g)(x)$.
9. Find the inverse of $f(x) = \dfrac{3x-2}{x+1}$.
10. Explain why $f(x) = x^2 - 6x + 5$ needs a restricted domain to have a function inverse, and state a valid restriction.
11. Given $f(x) = 2x - 5$, find $f^{-1}(f(8))$ without computing $f^{-1}$'s formula, and explain why your shortcut works.
12. Given $g(x) = 3x - 2$ and $(f \circ g)(x) = 9x^2 - 12x + 7$, find $f(x)$.
13. Given $f(x) = \dfrac{x}{x-1}$, find $f^{-1}(x)$ and verify $f^{-1}$ has the same rule as $f$ (i.e. $f$ is its own inverse).

## ✓ Answer Key

1. $f(2) = 8$, $g(8) = 16$
2. $g(x + 6) = 2x + 12$
3. $f^{-1}(x) = \dfrac{x+3}{5}$
4. $f^{-1}(x) = 3x - 4$
5. $y = -\sqrt{x-2}$ (negative root, matching $x \leq 0$)
6. $f(g(x)) = 2\left(\frac{x-6}{2}\right) + 6 = x$; $g(f(x)) = \frac{(2x+6)-6}{2} = x$: both confirm they're inverses
7. $(g \circ f)(3) = g(2) = 4$; $(f \circ g)(3) = f(9) = 8$ — different, showing composition order matters
8. $\sqrt{x+8}$ requires $x + 8 \geq 0$: domain $x \geq -8$
9. $x(y+1) = 3y - 2 \Rightarrow xy + x = 3y - 2 \Rightarrow y(x-3) = -x - 2 \Rightarrow y = \dfrac{-x-2}{x-3} = \dfrac{x+2}{3-x}$
10. vertex at $x = 3$; $f$ fails the horizontal line test around it; restrict to $x \geq 3$ (or $x \leq 3$)
11. $f^{-1}(f(8)) = 8$, since applying a function then its inverse always returns the original input
12. let $u = 3x - 2 \Rightarrow x = \frac{u+2}{3}$; substitute into $9x^2 - 12x + 7$, simplifies to $u^2 + 3$: $f(x) = x^2 + 3$
13. $y = \dfrac{x}{x-1}$; swap: $x = \dfrac{y}{y-1} \Rightarrow x(y-1) = y \Rightarrow xy - x = y \Rightarrow y(x-1) = x \Rightarrow y = \dfrac{x}{x-1}$ — same rule, so $f = f^{-1}$